

Hoyoung Kim

khy@dankook.ac.kr | COIS Lab

EDUCATION

Dankook University

M.S. in Industrial Engineering

Cheonan, Republic of Korea

Mar. 2026 – Present

- Computational Optimization for Intelligent Systems (COIS) Lab

Dankook University

B.S. in Management Engineering GPA: 4.09/4.50

Cheonan, Republic of Korea

Graduated Feb. 2026

EXPERIENCE

Dankook University

Researcher

Cheonan, Republic of Korea

Dec. 2024 – Present

- Conducted various public and private projects as a project manager in Computational Optimization for Intelligent Systems Laboratory at Dankook University.

SKILLS

Programming Languages: Python, SQL

Optimization: Gurobi

Technical Software: AutoMod, AutoCAD

Certifications: SQL Developer (SQLD)

PUBLICATIONS(DOMESTIC)

1. Han, G., Noh, Y., Woo, M., Hong, J., **Kim, H.**, Kim, H., Cho, S. W.[†] (2026). Social Network Analysis of Regional Health Recognition Inequality Factors: A Focus on Secondhand Smoke. *Health and Social Welfare Review*. (KCI)
2. **Kim, H.**, Kang, S., Seol, H., Yoo, C., Hong, B., Cho, S. W.[†] (2025). Patent-Driven Topic Modeling and Opportunity Analysis for Smart Logistics Technology. *Journal of Logistics Science & Technology*. (KCI)

CONFERENCES(INTERNATIONAL)

1. Woo, M., Kim, H., **Kim, H.**, Zhou, Y., Cho, S. W.[†] (2026). Neural Column Generation for Quay Crane Scheduling. *International Conference on Logistics and Maritime Systems (LOGMS)*, pp. 27–39.

CONFERENCES(DOMESTIC)

1. **Kim, H.**, Han, G., Kim, J., Woo, M., Kim, H., Lee, W., Cho, S. W.[†] (2026). 머신러닝 기반의 연료 소모량 예측을 활용한 선박 속도 최적화. *KASOTS 2026*, Jeju, Republic of Korea.
2. **Kim, H.**, Kang, S., Seol, H., Yoo, C., Hong, B., Cho, S. W.[†] (2025). 스마트 물류 기술 동향 분석을 위한 특허 데이터 기반 토픽모델링 및 기술 기획 분석. *LOGIST 2025*, Daegu, Republic of Korea.

PROJECTS

Data-Driven FOC Prediction & Speed Optimization

May 2026 – Nov. 2026

Korea Research Institute of Ships & Ocean Engineering (KRISO)

- Developed an ML-based fuel-oil consumption (FOC) prediction module accounting for real-world maritime environments, and formulated a MILP speed optimization algorithm to minimize total fuel usage across the entire voyage.

Credit Rating Model Development <i>NICE Information Service</i>	Sep. 2025 – Dec. 2025
Engineered automated financial and non-financial corporate credit rating models using multivariate statistical analysis, including a specialized framework for the shipping industry.	
AI Investment Model & Capability Modeling <i>Korea Technology Finance Corporation (KOTEC)</i>	Jul. 2025 – Dec. 2025
Modeled the interaction structure of business sectors using non-linear ML algorithms to estimate intrinsic innovation expectancy and corporate future growth potential.	
VDES Network Resource Allocation Optimization <i>Ministry of Oceans and Fisheries</i>	Feb. 2025 – Dec. 2027
Developed a hierarchical MIP model to optimize base station resources (frequency, TDMA, time slot selection) for VHF Data Exchange Systems (VDES).	
Dynamic Cell Planning (DCP) Algorithm Development <i>AIVeNautics</i>	Sep. 2025 – Nov. 2025
Formulated a mathematical optimization model to dynamically redesign AIS cell boundaries in real time, maximizing network coverage while incorporating satellite-internet characteristics such as Starlink.	
Timetable Optimization Service (TOS) Algorithm <i>Dankook University</i>	Jul. 2025
Developed a Gurobi-based scheduling optimization algorithm that reflects user preferences for undergraduate course planning.	

AWARDS AND HONORS

Top Excellence Award , LOGIST 2025 Logistics Paper Competition	Nov. 2025
Encouragement Award , 21st KMAC Management Innovation Competition	Nov. 2025
Beomeun Scholarship Foundation , Scholarship Student	Nov. 2025
Excellence Award , DKU CTL Best Practice Competition	Jul. 2025
Academic Excellence Awards , Dankook University (성적우수상, 우등상)	2022 – 2026